Solution: Maximize Capital

Problem Statement

Given a set of investment projects with their respective profits, we need to find the **most profitable projects**. We are given an initial capital and are allowed to invest only in a fixed number of projects. Our goal is to choose projects that give us the maximum profit. Write a function that returns the maximum total capital after selecting the most profitable projects.

We can start an investment project only when we have the required capital. After selecting a project, we can assume that its profit has become our capital, and that we have also received our capital back.

Example 1:

```
Input: Project Capitals=[0,1,2], Project Profits=[1,2,3], Initial
Capital=1, Number of Projects=2

Output: 6

Explanation:
1. With initial capital of ‘1’, we will start the second project which
will give us profit of ‘2’. Once we selected our first project, our
total capital will become 3 (profit + initial capital).
2. With ‘3’ capital, we will select the third project, which will give
us ‘3’ profit.

After the completion of the two projects, our total capital will be 6
(1+2+3).
```

Example 2:

```
Input: Project Capitals=[0,1,2,3], Project Profits=[1,2,3,5], Initial
Capital=0, Number of Projects=3

Output: 8

Explanation:

1. With ‘0’ capital, we can only select the first project, bringing out
capital to 1.
2. Next, we will select the second project, which will bring our capital
to 3.
3. Next, we will select the fourth project, giving us a profit of 5.

After selecting the three projects, our total capital will be 8 (1+2+5).
```

Solution

While selecting projects we have two constraints:

1. We can select a project only when we have the required capital.
2. There is a maximum limit on how many projects we can select.

Since we don’t have any constraint on time, we should choose a project, among the projects for which we have enough capital, which gives us a maximum profit. Following this greedy approach will give us the best solution.

While selecting a project, we will do two things:

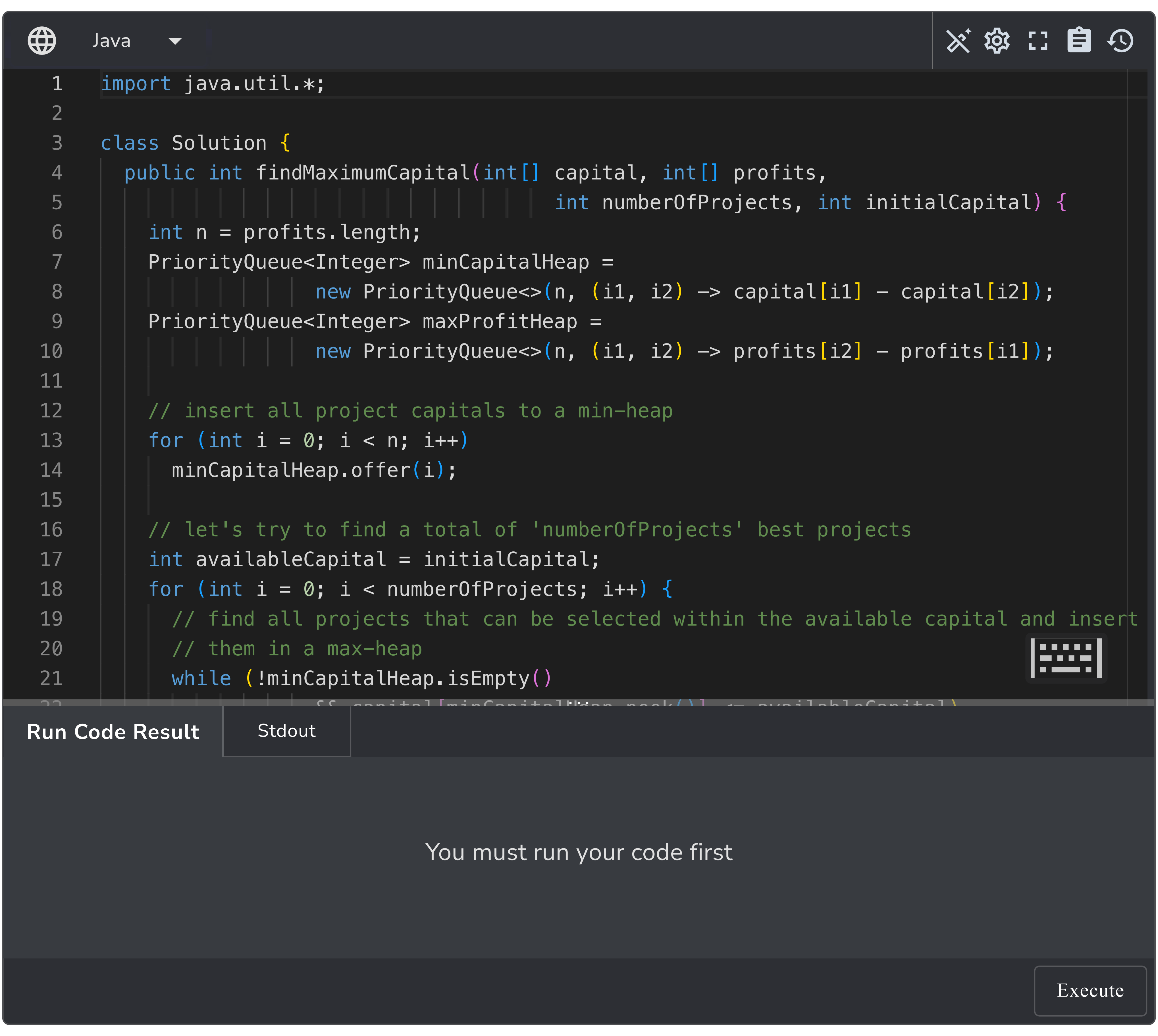
1. Find all the projects that we can choose with the available capital.
2. From the list of projects in the 1st step, choose the project that gives us a maximum profit.

We can follow the **Two Heaps** approach similar to **Find the Median of a Number Stream**. Here are the steps of our algorithm:

1. Add all project capitals to a min-heap, so that we can select a project with the smallest capital requirement.
2. Go through the top projects of the min-heap and filter the projects that can be completed within our available capital. Insert the profits of all these projects into a max-heap, so that we can choose a project with the maximum profit.
3. Finally, select the top project of the max-heap for investment.
4. Repeat the 2nd and 3rd steps for the required number of projects.

Code

Here is what our algorithm will look like:



```java
import java.util.*;

class Solution {
  public int findMaximumCapital(int[] capital, int[] profits,
                                int numberOfProjects, int initialCapital) {
    int n = profits.length;
    PriorityQueue<Integer> minCapitalHeap =
                  new PriorityQueue<>(n, (i1, i2) -> capital[i1] - capital[i2]);
    PriorityQueue<Integer> maxProfitHeap =
                  new PriorityQueue<>(n, (i1, i2) -> profits[i2] - profits[i1]);

    // insert all project capitals to a min-heap
    for (int i = 0; i < n; i++)
      minCapitalHeap.offer(i);

    // let's try to find a total of 'numberOfProjects' best projects
    int availableCapital = initialCapital;
    for (int i = 0; i < numberOfProjects; i++) {
      // find all projects that can be selected within the available capital and insert
      // them in a max-heap
      while (!minCapitalHeap.isEmpty()
```

You must run your code first

Time Complexity

Since, at the most, all the projects will be pushed to both the heaps once, the time complexity of our algorithm is O(NlogN + KlogN), where ‘N’ is the total number of projects and ‘K’ is the number of projects we are selecting.

Space Complexity

The space complexity will be O(N) because we will be storing all the projects in the heaps.

← Previous
Maximize Capital (Hard)
Next →
Problem Challenge 1: Next Interval (Hard)

☐ Mark as Completed